

Modeling Heat Transfer Correlations in Plate Heat Exchangers

Group B11

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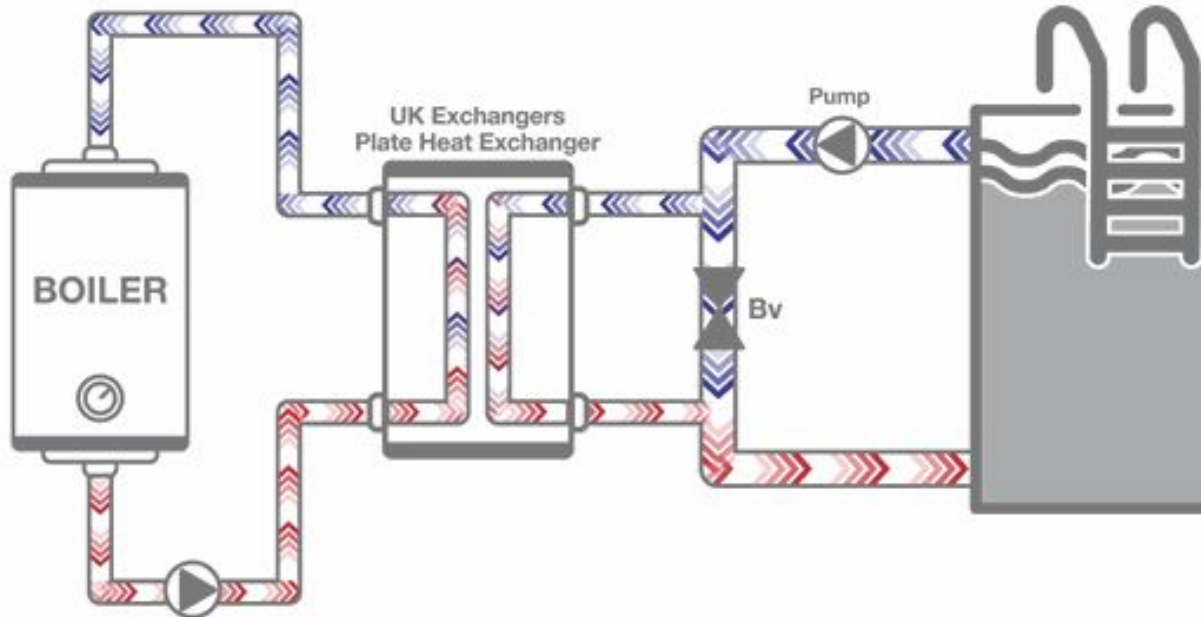
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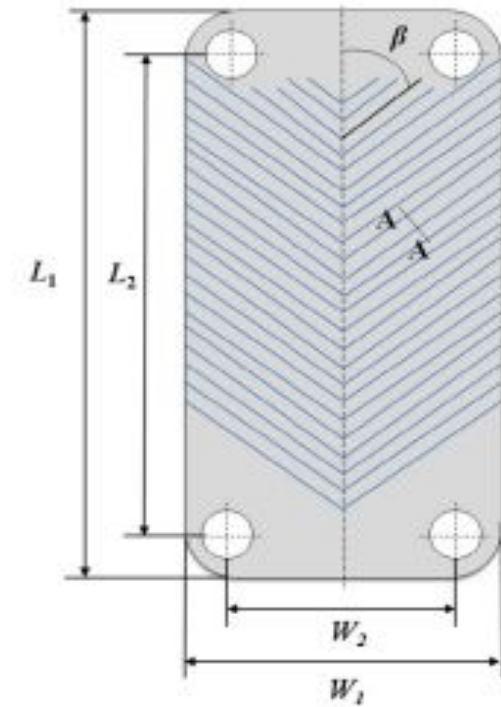


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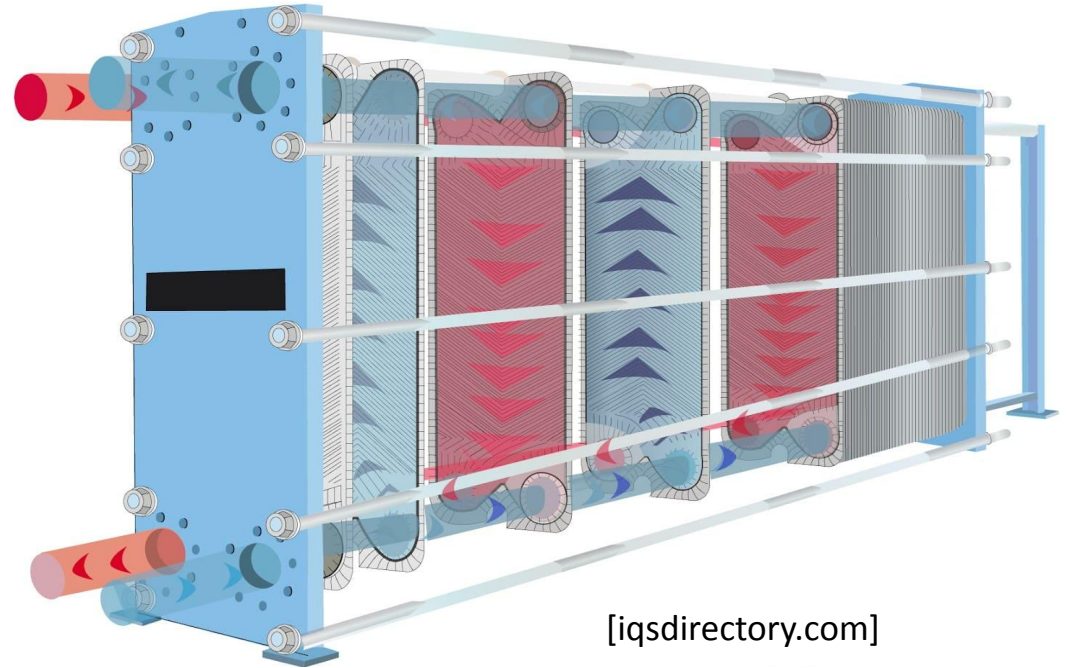
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[www.uk-exchangers.com]

Due to maximized heat transfer area, flat plate heat exchangers are extremely efficient.

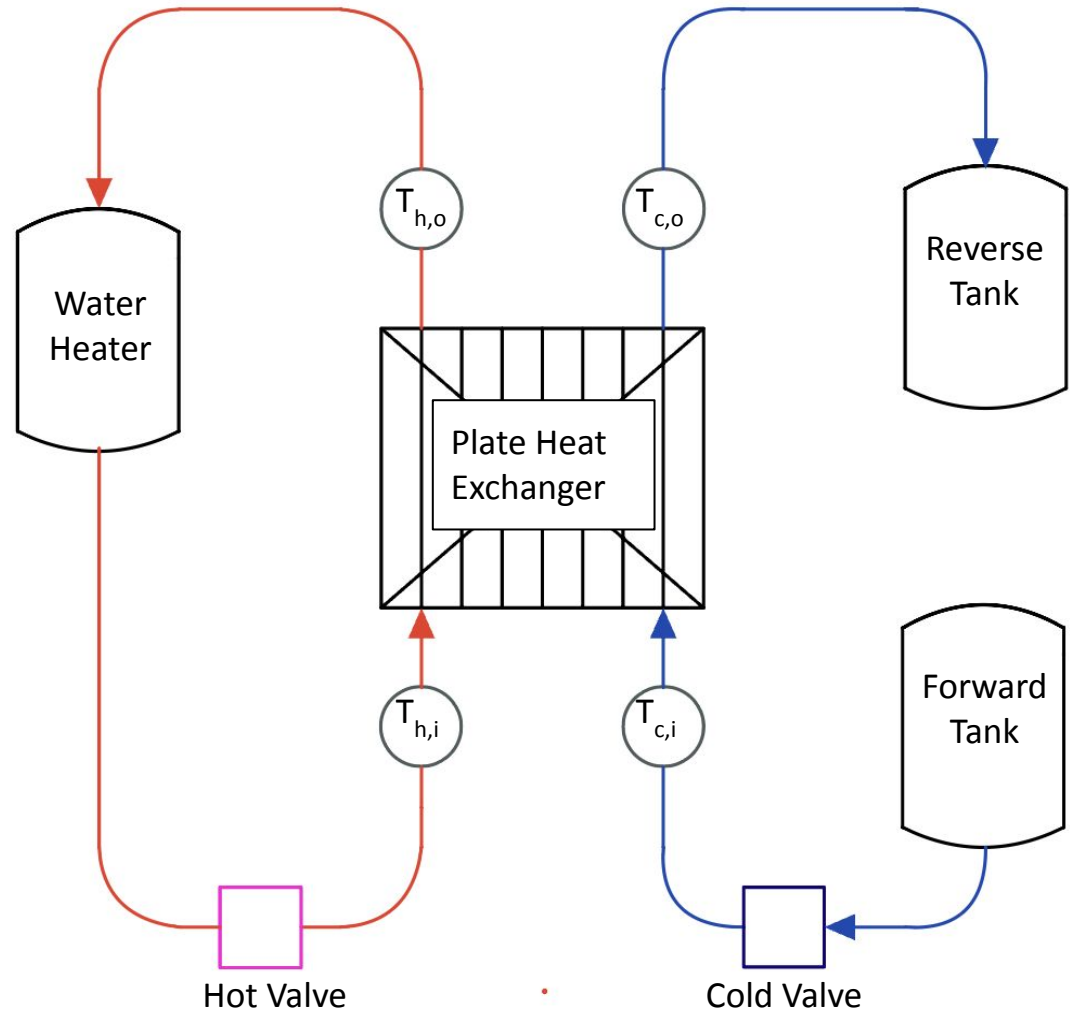
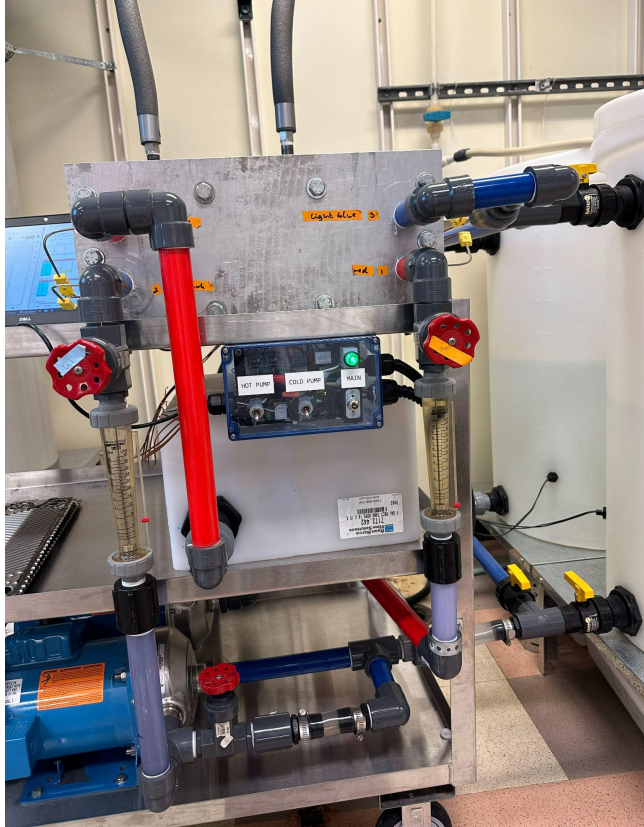


[Yang et al., 2017]



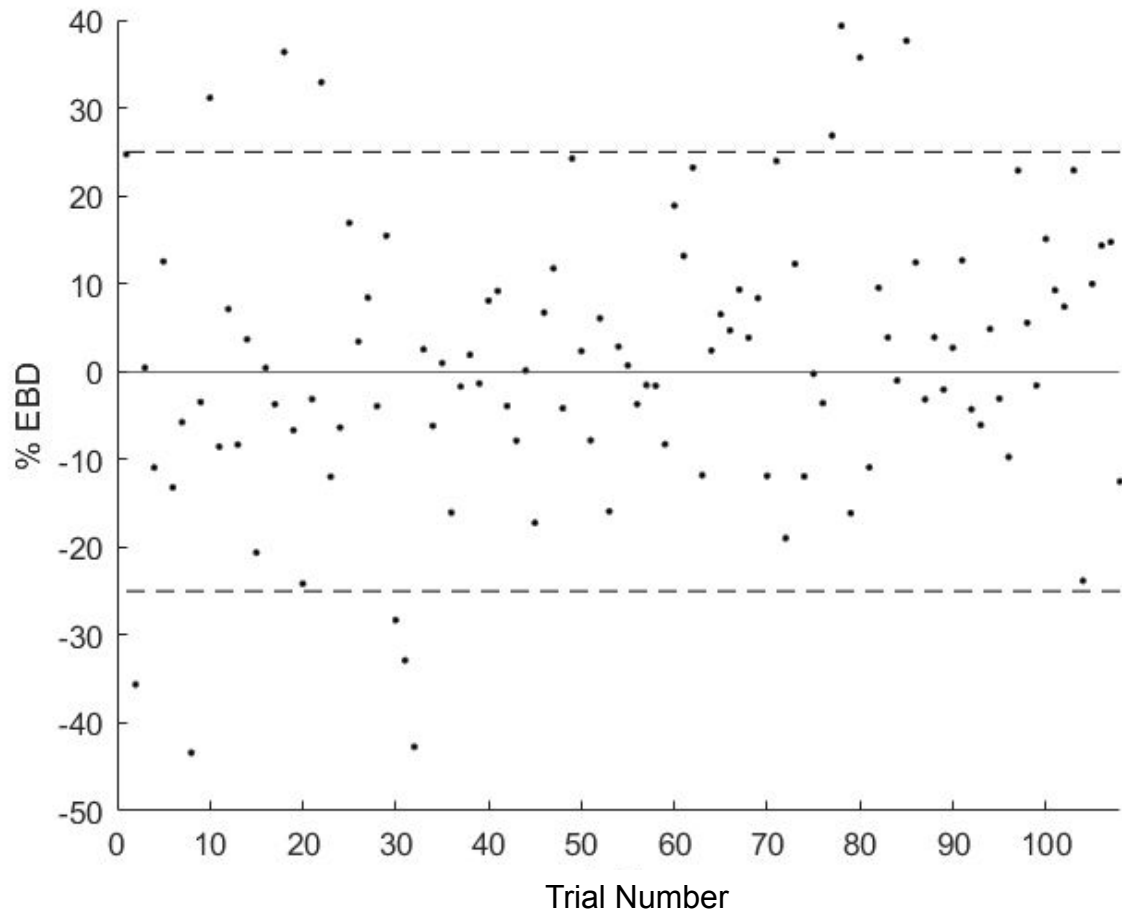
[iqsdirectory.com]

Experimental temperature data was obtained by varying volumetric flow rate of hot and cold streams.



Energy Balance Deviation (EBD) suggests that sufficient heat transfer was observed between streams.

$$EBD \% = \frac{Q_h - Q_c}{Q_{avg}}$$

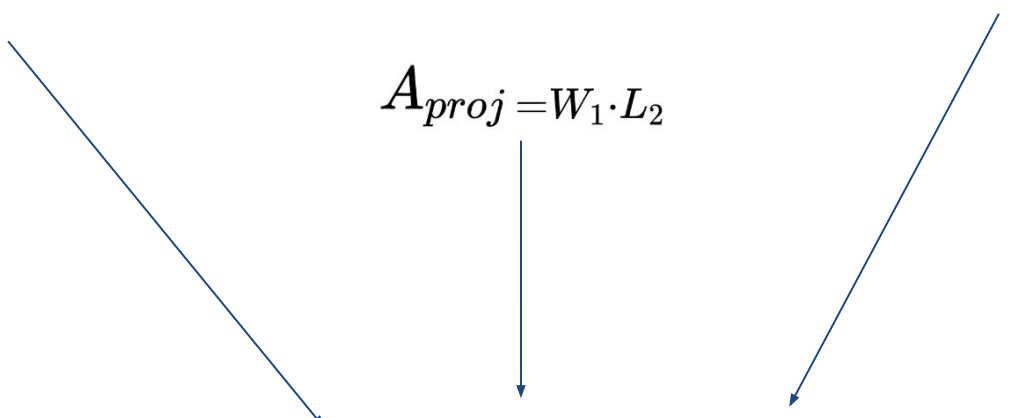


Measured overall heat transfer coefficient was determined using experimental data

$$Q = \frac{m_h c_h (T_{h,i} - T_{h,o}) + m_c c_c (T_{c,o} - T_{c,i})}{2}$$

$$\Delta T_{LMTD} = \frac{(T_{h,i} - T_{c,o}) - (T_{h,p} - T_{c,i})}{\ln[(T_{h,i} - T_{c,o}) - (T_{h,p} - T_{c,i})]}$$

$$A_{proj} = W_1 \cdot L_2$$


$$U_{meas} = \frac{Q}{A_{proj} \cdot \Delta T_{LMTD}}$$

Calculated overall heat transfer coefficient was determined using a nonlinear fit in MATLAB.

$$Nu = C_1 Re^{C_2} Pr^{\frac{1}{3}}$$

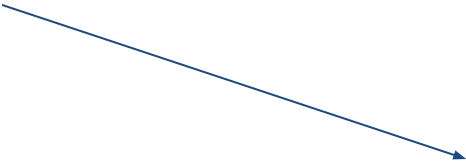

$$\begin{bmatrix} C_1 \\ C_2 \end{bmatrix} = nlinfit\left(X, U_{meas}, @CalcU, \begin{bmatrix} C_{1,guess} \\ C_{2,guess} \end{bmatrix}\right)$$

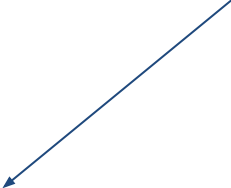

$$C_1 = 1.7868$$

$$C_2 = 0.6853$$

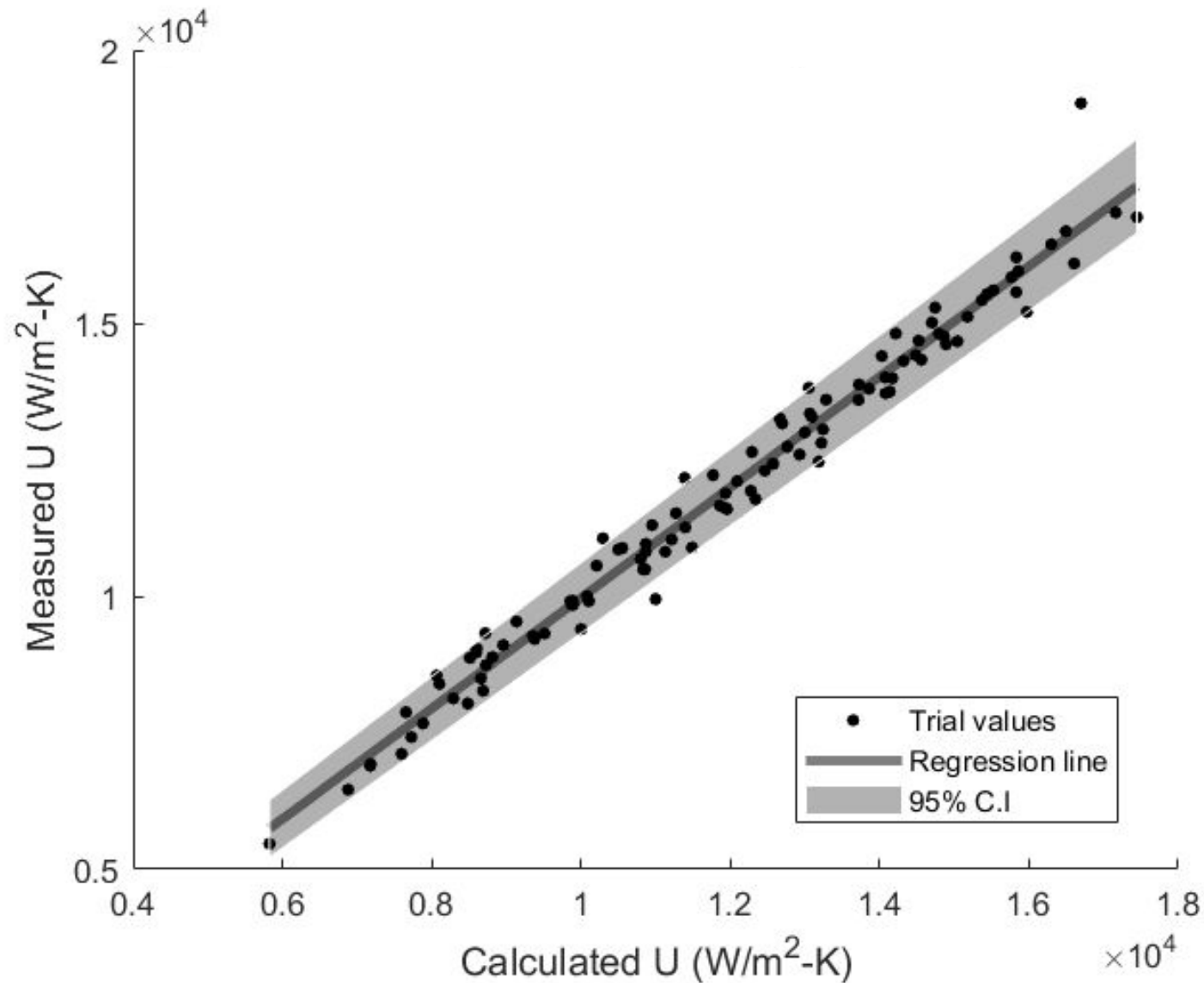
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$$Nu = C_1 Re^{C_2} Pr^{\frac{1}{3}}$$


$$h = \frac{Nu \cdot k}{D_h}$$


$$\frac{1}{U_{calc}} = \frac{1}{k} + \frac{1}{h_c} + \frac{1}{h_h}$$

U_{meas} and U_{calc} display a near-linear relationship.



In conclusion, the experimental findings are similar to expected theoretical values.

This model could be used to characterize other plate heat exchangers, assuming that both water streams remain within the given temperature range.



[hrs-heatexchangers.com]

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Questions?

Citations:

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<https://www.iqsdirectory.com/articles/heat-exchanger/plate-heat-exchangers/basic-structure-of-a-plate-heat-exchanger.jpg>

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